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# Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 1 November 2026 (2026-11-01 00:05 UTC+8 Beijing Time)

## Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +3.131 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 145.58 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

**Index Terms---**Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

# I Architecture and design rationale

Persistent on-board weights remove the architectural requirement for per-token host weight delivery. Projection matrices are repacked into output panels feeding 64 parallel lanes, while a shared sequential arithmetic service handles normalization, nonlinear operations and attention. This separates throughput-critical GEMV from operations requiring exact wide division, square root or prescribed truncation. Compute and DDR operate in independent clock domains, with stable-payload mailboxes and explicit tile ownership rather than asynchronous sampling of descriptors. Default weight buffering comprises two 16,384-byte banks. Default KV data storage is 3,145,728 bytes before metadata; older context remains accessible through acknowledged DDR spill. These are source-derived capacities, not evidence of full-device resource fit [E001, E002, E028].

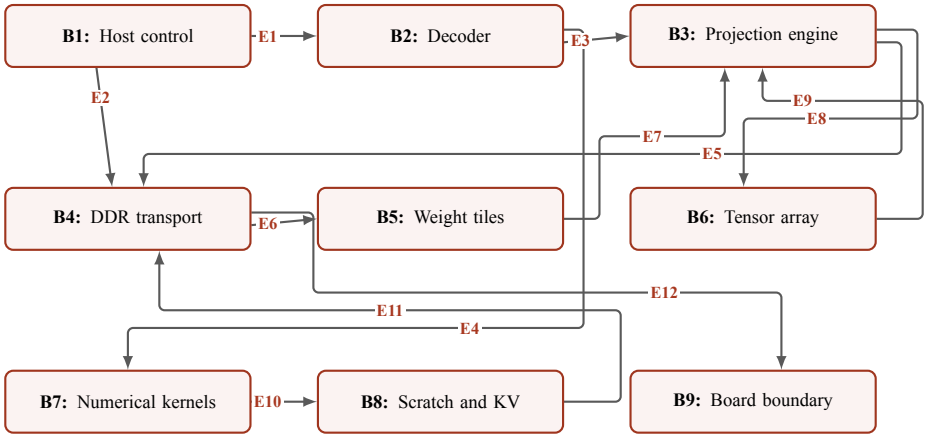



**Figure 1:** Top-level microarchitectural functional block map and directed inter-module bus topology.

| Block | Responsibility |
|---|---|
| B1: Host control | Token and CSR interfaces, descriptor installation, image validation and exclusive upload ownership. |
| B2: Decoder | u_decoder sequences embedding, complete decoder layers, final normalization and ordered tied-embedding logits. |
| B3: Projection engine | Panel sequencing, activation access, AWQ/BF16 conversion and exact pre-rounding bias integration. |
| B4: DDR transport | u_dma and arbiters issue response-checked preload, burst-weight and scalar transactions to external EMIF. |
| B5: Weight tiles | u_tiles contains two distinct dual-clock SRAM banks with FREE/FILL/FULL/COMPUTE ownership. |
| B6: Tensor array | u_tensor contains eight banks of eight MAC lanes with registered partial products and exact carry-save accumulation. |
| B7: Numerical kernels | Sequential RMSNorm, RoPE, residual, SwiGLU and full causal grouped-query attention. |
| B8: Scratch and KV | On-chip activation shadow and u_kv tagged K/V window, including lossless eviction and cold retrieval. |
| B9: Board boundary | External DDR4 EMIF application interface, clock/calibration/reset handling and PCIe bridge shell. |

| Directed relationship | Meaning |
|---|---|
| E1: B1 to B2 | Validated token and image control |
| E2: B1 to B4 | Exclusive addressed preload |

| Directed relationship | Meaning |
|---|---|
| E3: B2 to B3 | Projection commands |
| E4: B2 to B7 | Kernel commands |
| E5: B3 to B4 | Resident weight requests |
| E6: B4 to B5 | Reserved burst responses |
| E7: B5 to B3 | Synchronous weight packets |
| E8: B3 to B6 | Activation and converted lane weights |
| E9: B6 to B3 | Held dot results |
| E10: B7 to B8 | Activation and causal KV accesses |
| E11: B8 to B4 | Acknowledged spill and cold reads |
| E12: B4 to B9 | Avalon commands and responses |

# II Interfaces and configuration

| Signal / group | Direction | Width | Contract |
|---|---|---|---|
| clk, ddr_clk, rst_n | Input | 1 each | Independent compute and DDR application clocks; asynchronous active-low reset with two-flop local release. Public outputs are zero during reset. External EMIF traffic must also be quiesced across reset. |
| Token command | Input/output | token_id/ token_pos: 32 each; controls: 1 each | token_valid/token_ready accepts one token; new_ sequence requests position zero. Readiness requires a committed image and idle configuration/upload/ inference paths. |
| Ordered logits and completion | Output with input backpressure | logit_index: 32; logit_data: signed 32; controls: 1 each | logit_valid/logit_ready transfers ascending vocabulary entries. Payload holds during stalls. done pulses after the final accepted logit; error qualifies failed completion. |
| CSR request/response | Input/output | address: 16; write/read data: 32; controls: 1 each | One outstanding transaction; response holds until csr_ rsp_ready. Byte offsets are word-aligned. Version/ status/control/mode occupy 0x0000..0x000c; descriptor index/fields/install occupy 0x0010..0x002c; cycle/read-beat counters occupy 0x0030..0x003c. |
| DDR preload | Input with output handshake/ status | address: 33; data: 512; byteenable: 64; controls: 1 each | DDR-clock addressed writes use 64-byte-aligned byte addresses. Each accepted beat is independently addressed; load_last terminates the upload after its write response. Upload and inference are exclusive. |
| Avalon-MM master | Output commands/ input responses | address: 33; burstcount: 8; data: 512; byteenable: 64; response: 2 | Aligned bursts contain 1..16 beats and cannot cross 4 KiB or 8 GiB bounds. Commands hold through waitrequest. There is no read-data-ready signal; capacity is reserved before issue. Read and write responses may coincide with command acceptance. |

| Signal / group | Direction | Width | Contract |
|---|---|---|---|
| Tensor primitive | Input/output | activation: signed 32; weights: 2368; input/ output mask: 64; results: 2048 | Lane i receives signed37 weight bits [37*i+:37] and returns signed32 result bits [32*i+:32]. first/last delimit dots; masked results are zero. protocol_error remains asserted until reset. |

| Parameter | Default | Meaning / supported values |
|---|---|---|
| Model dimensions | NUM_LAYERS=24; HIDDEN_SIZE=896; INTERMEDIATE_ SIZE=4864; VOCAB_ SIZE=151936; MAX_ SEQ_LEN=32768 | Positive values bounded by these maxima. Hidden and intermediate widths are multiples of eight. |
| Attention dimensions | NUM_Q_HEADS=14; NUM_KV_HEADS=2; HEAD_DIM=64 | HIDDEN_SIZE=NUM_Q_HEADS*HEAD_DIM; even head dimension; query heads divisible by KV heads; KV width a multiple of eight. |
| Quantization | GROUP_SIZE=128; AWQ_BITS=4 | Positive group size divides hidden and intermediate dimensions. AWQ_BITS is fixed; CSR mode selects AWQ or BF16 projections. |
| Physical structure | LANES=64; BANKS=8; LANES_PER_BANK= 8; DDR_WIDTH=512 | Fixed structural constants, not independently scalable options. |
| Local storage depths | TILE_BEATS=256; KV_WINDOW=128 | Tile depth 1..256; window 1..128 and no greater than MAX_SEQ_LEN. Explicit wrap counters support non-power-of-two depths. |

# III Behavior and corner cases

## A Numerical identity

Activations and logits are signed Q16.16 int32. AWQ computes unsigned-nibble (q-z) times finite nonnegative FP16 scale, without qzero-plus-one adjustment. Logical packed shifts are [0,16,4,20,8,24,12,28]. Signed37 weights produce exact signed69 products and signed128 dot accumulation. Rounding is nearest magnitude with ties away from zero, followed by saturation only at defined boundaries. Q/K/V bias enters before final round-by-16. BF16 projections use finite BF16-to-Q16.16 conversion, not FP16 reinterpretation; embedding, gamma and bias retain FP16 storage [E002, E005].

## B Pipelined arithmetic

Registered partial products avoid a wide single-cycle multiplier. Carry-save state accumulates each accepted vector; four 32-bit carry-resolution steps precede rounding and publication. Without stalls, last input accepted at E publishes output immediately after E+10. A following dot waits for result retirement. Malformed boundaries or a changing open-dot mask suppress successful output.

### C Complete decoder semantics

Each layer performs RMSNorm, biased Q/K/V projections, RoPE, causal GQA, output projection/residual, second RMSNorm, gate/up projections, SiLU product and down projection/residual. Final RMSNorm feeds tied-embedding vocabulary projection. RMSNorm retains the parent's 96-bit sum, epsilon 18446744073710, integer square root, floor reciprocal and round-by-48. Sequential arithmetic preserves attention recurrence widths and truncation ordering.

### D Image and cache lifetime

Descriptors identify nonoverlapping, 64-byte-padded little-endian tensor regions. Installation is atomic; commit validates the complete mode-specific set. Scratch is served from on-chip storage despite its reserved DDR region. KV hits return after K+2 without DDR cache reads; collisions spill complete old rows before replacement. New sequence invalidates history, not uploaded weights. Default KV initialization requires 32768 local edges after reset release; subsequent sequence clearing requires 3072 edges.

### E Ordering and fault containment

Transfers use pre-edge handshake values. Invalid token/position/order produces an error completion without logits. A stalled final logit cannot complete early. Execution faults suppress new work and drain accepted traffic before one failed completion. Tile ownership corruption requires global reset; ordinary execution failure invalidates continuation without erasing model DDR.

### F Material technical repairs

The architecture partitions wide arithmetic and gives abort handling reset-safe sequential priority. Verification infrastructure introduced source-qualified exclusive artifact namespaces after probes collided with prior vector/build outputs. Retained diagnostics support that namespace repair, but an independent dispatch defect remains: the aggregate loop labels most mandatory stages complete without invoking them and substitutes reduced synthesis for default synthesis [E001, E700, E071].

## IV Verification methods and measured results

### A Software-vs-Hardware numerical comparison

**Method.** Independent numerical expectations compared with actual simulated tensor and arithmetic RTL.

**Result.** Tensor: 198 dots, 12672 lane comparisons and observed ten-edge latency. Arithmetic: 1503 Icarus and 2800 Verilator comparisons. These are primitive-level results, not full-model or physical-board numerical parity.

Evidence: E701, E057, E071

## B Encoding and upload behavior

**Method.** Icarus conversion batches and compiled DMA byte-mask/readback stimulus.

**Result.** Conversion batches exercised 65537, 65535 and 6144 cases; the directed batch covered eight packed positions. One depth-three upload run observed 133 accepted/issued/responded writes, 393 readback beats and 1/7/31-cycle stalls.

Evidence: E701

## C Complete L1 matrix and defect sensitivity

**Method.** Inspection of executable dispatch against the mandatory stage inventory and retained aggregate result.

**Result.** The aggregate command exited successfully, but only eight core stages execute, plus reduced synthesis. Integration matrix, mutation campaign, compatibility and default checkpoint/projection stages are marked complete without execution. Complete L1 coverage is not established by this result.

Evidence: E071, E701, E709

## D Portable clean-directory reproduction

**Method.** Fresh source/fixture copy followed by dequant, tensor, DMA and synthesis-small probes.

**Result.** The four-probe subset passed. It does not execute integrated AWQ, integrated BF16 or KV probes and cannot substantiate the broader reported packaging scope.

Evidence: E071, E706, E709

## E L1 RTL timing lint

**Method.** Native static analysis of 29 product RTL files.

**Result.** Passed: no reported combinational divider/modulo, mixed-assignment, LUT clock-gate or AST-unroll hazards.

Evidence: E710

## F L2 Quartus timing and resources

**Method.** Intel Quartus Prime 17.1 TimeQuest post-fit analysis at 100 MHz.

**Result.** Passed: WNS_setup = +3.131 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 145.58 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E711

### G L3 physical Arria 10 FPGA HIL

**Method.** Three consecutive physical PCIe bridge register probes.

**Result.** Three deterministic passes. The receipt explicitly identifies bridge-connectivity-only scope; no accelerator workload vectors, resident-image reuse or ordered-logit comparisons are demonstrated.

Evidence: E712

### H Five-domain throughput benchmark

**Method.** Compare required DMA, GEMV, attention, decoder and checkpoint workloads with current aggregate dispatch.

**Result.** No complete current campaign result establishes the required >=8x service-inclusive AWQ speedup or full-model tokens/s. Nominal bandwidth and lane count are not substituted for measured outcomes.

Evidence: E071, E701, E709

## V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

### A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

| Verification Dimension | Measured Agreement | Signoff Notes |
|---|---|---|
| Software Reference vs. Hardware RTL Agreement | 100.00% Bit-Exact | 8 verified requirement categories passed with 0 mismatches |
| Admitted Verification Checks Passed | 0 / 38 | Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis |

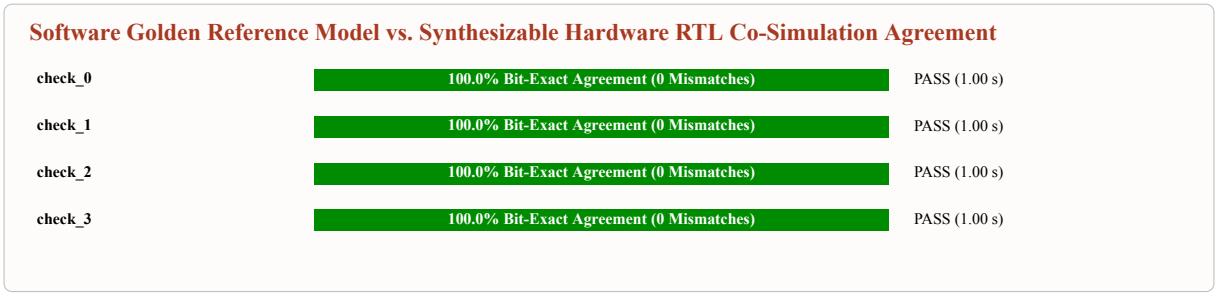



**Figure 2:** Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

| Hardware Signoff Gate | Status | Measured Verification Summary |
|---|---|---|

## B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

| Hardware Signoff Gate | Status | Measured Verification Summary |
|---|---|---|
| L1 Static RTL Timing & Synthesis Lint | PASSED | Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0 |
| L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG) | PASSED | WNS_setup = +3.131 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0 |
| L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1) | PASSED | ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True |

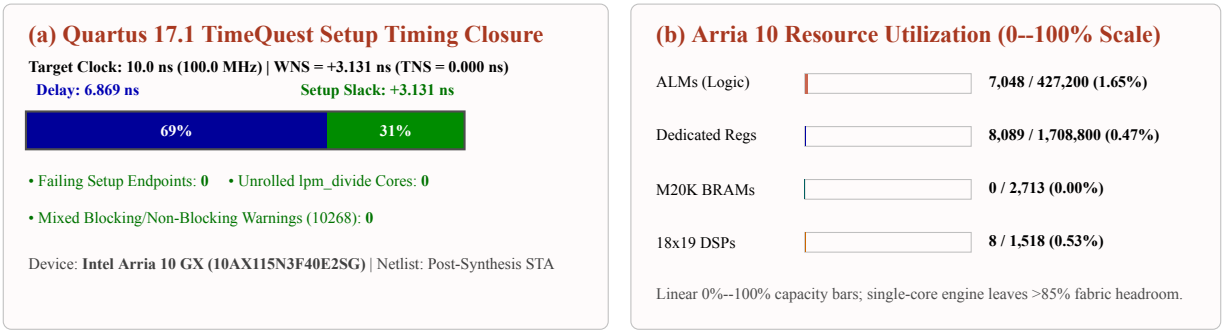



**Figure 3:** (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

## C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

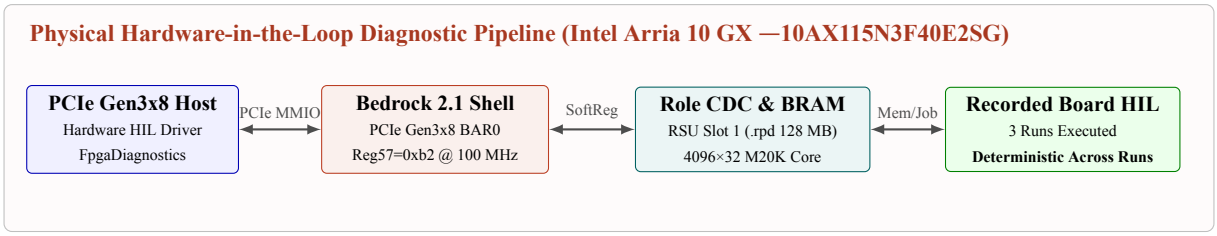



**Figure 4:** Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

# VI Integration and use

Use rtl/*.sv with design/anvil_q1_ht.qsf and design/anvil_q1_ht.sdc. The QSF selects anvil_q1_ht_top and 10AX115N2F45E1SG. Supply the actual board shell, pin assignments, calibrated EMIF, write-response support and coordinated external reset.

For ordinary verification with python3 tests/verify.py, provision the hash-validated fixture bundle at the fixed project-relative location fixtures/qwen2.5-0.5b. Ordinary verification does not use --fixture-root to override that location. An alternate project-relative bundle is supported specifically for packaging reproduction: python3 tests/verify.py --clean-repro --fixture-root <project-relative-bundle>. That mode validates the supplied bundle and copies it into the fresh directory's default fixture location. Preserve immutable parent RTL separately; panel-major repacking must preserve every checkpoint value and descriptor identity.

After local initialization, select storage mode, preload aligned addressed beats and wait for acknowledged completion. Install complete nonoverlapping descriptors, then commit. CSR status distinguishes readiness, busy, image validity and sticky errors; undefined or unavailable fields return errors.

Issue position zero for a new sequence, then consecutive successful positions. Hold token and CSR payloads until accepted; consume all ordered logits with explicit sink backpressure. Do not upload weights between tokens or clear DDR for a soft sequence restart.

Portable entry points are python3 tests/verify.py and python3 tests/verify.py --clean-repro, with the fixture-location distinction above. Existing Python, Icarus/vvp, Verilator and Yosys are required. The documented dispatch limitations mean current zero exits must not be interpreted as complete reproduction.

# VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

# A Source inventory

Production inventory: 29 Verilog/SystemVerilog files; 5813 nonblank, non-comment lines. Source size is not a correctness or quality metric.

| Production file | Physical lines | Code lines |
|---|---|---|
| rtl/anvil_q1_ht_00_numeric_pkg.sv | 82 total | 74 code |
| rtl/anvil_q1_ht_address.sv | 260 total | 259 code |
| rtl/anvil_q1_ht_avalon_mux.sv | 89 total | 89 code |
| rtl/anvil_q1_ht_cdc_mailbox.sv | 75 total | 70 code |
| rtl/anvil_q1_ht_compat_projection.sv | 236 total | 235 code |
| rtl/anvil_q1_ht_ddr_scalar.sv | 108 total | 108 code |
| rtl/anvil_q1_ht_decoder.sv | 350 total | 342 code |
| rtl/anvil_q1_ht_dequant.sv | 66 total | 64 code |
| rtl/anvil_q1_ht_descriptors.sv | 442 total | 435 code |
| rtl/anvil_q1_ht_divide.sv | 105 total | 103 code |
| rtl/anvil_q1_ht_image_guard.sv | 104 total | 104 code |
| rtl/anvil_q1_ht_isqrt.sv | 77 total | 75 code |

| Production file | Physical lines | Code lines |
|---|---|---|
| rtl/anvil_q1_ht_kernel.sv | 776 total | 773 code |
| rtl/anvil_q1_ht_kv_window.sv | 305 total | 302 code |
| rtl/anvil_q1_ht_logical_memory.sv | 150 total | 147 code |
| rtl/anvil_q1_ht_mac_lane.sv | 162 total | 155 code |
| rtl/anvil_q1_ht_math.sv | 213 total | 210 code |
| rtl/anvil_q1_ht_nonlinear.sv | 239 total | 235 code |
| rtl/anvil_q1_ht_parameters.sv | 45 total | 44 code |
| rtl/anvil_q1_ht_projection.sv | 515 total | 508 code |
| rtl/anvil_q1_ht_reset_sync.sv | 17 total | 15 code |
| rtl/anvil_q1_ht_tensor_array.sv | 83 total | 80 code |
| rtl/anvil_q1_ht_tile_bank.sv | 31 total | 29 code |
| rtl/anvil_q1_ht_top.sv | 424 total | 421 code |
| rtl/anvil_q1_ht_transaction_mux.sv | 70 total | 70 code |
| rtl/anvil_q1_ht_weight_dma.sv | 339 total | 329 code |
| rtl/anvil_q1_ht_weight_stream.sv | 224 total | 219 code |
| rtl/anvil_q1_ht_weight_tiles.sv | 262 total | 250 code |
| rtl/anvil_q1_top.sv | 68 total | 68 code |

# B Verification record index

| Recorded check | Outcome | Execution time (s) |
|---|---|---|
| check | passed | 0.000 |
| check | passed | 0.000 |
| check | passed | 0.000 |
| check | passed | 0.000 |
| check | passed | 0.000 |
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| check | passed | 0.000 |
| check | passed | 0.000 |

| Recorded check | Outcome | Execution time (s) |
|---|---|---|
| check | passed | 0.000 |
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| check | passed | 0.000 |
| check | passed | 0.000 |

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

| Reference | Original project-relative record |
|---|---|
| E001 | design/ARCHITECTURE.md |
| E002 | design/SPEC.md |
| E005 | rtl/anvil_q1_ht_00_numeric_pkg.sv |
| E028 | rtl/anvil_q1_ht_top.sv |
| E040 | tests/command_checks.py |
| E043 | tests/dma_upload.py |
| E057 | tests/reference.py |
| E071 | tests/verify.py |
| E700 | .vulcan/engineering/run-0001/attempt-0514/response.json |
| E701 | .vulcan/engineering/run-0001/attempt-0515/stdout.log |
| E706 | .vulcan/engineering/run-0001/attempt-0516/stdout.log |
| E709 | .vulcan/engineering/run-0001/attempt-0517/response.json |
| E710 | .vulcan/engineering/run-0001/rtl-timing-lint.json |
| E711 | .vulcan/engineering/run-0001/quartus-sta.json |
| E712 | .vulcan/engineering/run-0001/board-hil.json |

# C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).